Calibration Impact on Comfort Scoring

Before / After Comparison on the Full 31-Recording Corpus

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This note reports the per-scenario impact of the calibration pipeline on the deployed comfort score. *Before* = hand-set default configuration (`config/default.yaml`); *After* = the calibrated configuration promoted to `config/deploy.yaml` after race 2 plus the no-pose ablation. Numbers are aggregated from a leave-one-recording-out (LORO) replay across all 31 RealSense `.bag` recordings; mean comfort and slope are computed over the intent + execution window per recording. Slope sign agreement is the fraction of recordings whose comfort slope matches the desired sign (`sc02`: positive; all others: negative).

Table 1: Per-scenario comfort metrics across all 31 LORO replays, before and after calibration. Numbers in **bold** highlight where calibration materially improved an entry against the desired behavior.

Scenario	n	Label	Mean comfort		Slope (pts/s)		Direction agree	
			Before	After	Before	After	Before	After
sc01_walkby	7	abort	69.2	50.2	−0.27	−0.39	5/7	5/7
sc02_comfortable	8	continue	68.3	**92.7**	−2.51	+0.73	0/8	**7/8**
sc03_gradual_disc.	5	abort	78.8	79.5	+0.16	−2.29	1/5	**5/5**
sc04_sudden_wdraw	5	abort	74.8	78.8	−0.96	−2.46	5/5	5/5
sc05_distracted	6	abort	73.2	73.0	−1.25	−3.28	6/6	6/6
Direction-correct total							17/31 (55%)	**28/31 (90%)**
Held-out test best Youden J (n=8)			0.00	**1.00**				
sc02 − sc01 mean gap			−0.9	+42.5				

Reading the table. Three things change under calibration. First, the *ordering* between abort and continue scenarios flips from inverted to a clean +42.5-point separation, driven entirely by `sc02` climbing from 68.3 to 92.7 and `sc01` falling from 69.2 to 50.2. Second, the *direction* of the comfort trajectory becomes correct on 28 of 31 recordings (90%) versus 17 of 31 (55%) under default — the largest gain is on `sc02`, where 0 of 8 successful handovers trended the right way under default, and 7 of 8 do under deploy. Third, `sc03` (gradual discomfort) flips from trending slightly *upward* (slope +0.16, wrong direction) to a strongly negative slope (−2.29), with direction agreement going from 1/5 to 5/5.

What didn't change. `sc04` (sudden withdrawal) and `sc05` (distracted) were already at 100% direction agreement under default; calibration steepens their downward slopes (more decisive aborts) but leaves the categorical classification unchanged. `sc01` (walk-by) direction agreement stays at 5/7 — the two outlier recordings remain ambiguous — but the absolute mean comfort drops 19 points, well below any plausible abort threshold, so the categorical decision is correct even on the disagreeing recordings.

The headline number. Best Youden's J on the held-out test set (n=8, including 3 aborts) goes from 0.00 at any threshold $\tau \in [30, 80]$ under default to 1.00 across the plateau $\tau \in [78, 80]$ under deploy (deployed at the conservative upper edge $\tau^* = 80$). The default configuration cannot separate abort from continue *at any threshold* in the swept range, while deploy clears the plateau by a wide margin — the two held-out sc02 recordings score 91.6 and 94.0, well above τ^*.

Optimization Search Space

Calibration tunes a 20-dimensional parameter vector via differential evolution against the cached parquets extracted in Stage A. Pretrained backbone weights (RetinaFace, EfficientNet-B0, L2CS-Net, MediaPipe Pose) are frozen; only the fusion / decision-layer parameters listed in Table 2 are searched. The abort threshold τ^* is selected post-hoc from the held-out ROC curve and is not part of the DE search. Source of truth: `PARAM_NAMES` and `BOUNDS` in `scripts/optimize_params.py`.

Table 2: Variables tweaked during few-shot optimization. Twenty parameters spanning fusion weights, sub-score coefficients, gaze/posture event thresholds and penalties, temporal smoothing, and missing-detection EMA decay. Bounds were chosen to keep DE candidates physically interpretable without artificially clipping the search.

Group	Parameter	Bounds	Role
Phase-aware fusion weights	`we_intent`	[0.1, 0.9]	emotion weight, intent phase
	`wp_intent`	[0.1, 0.9]	posture weight, intent phase
	`wg_intent`	[0.0, 1.0]	gaze weight, intent phase
	`we_exec`	[0.1, 0.9]	emotion weight, execution phase
	`wp_exec`	[0.1, 0.9]	posture weight, execution phase
	`wg_exec`	[0.0, 1.0]	gaze weight, execution phase
Emotion sub-score coefficients	`gamma`	[0.0, 1.0]	valence coefficient γ
	`delta`	[0.0, 1.0]	arousal coefficient δ
Gaze cone thresholds	`yaw_threshold`	[5°, 40°]	$\lvert\psi\rvert <$ this $\Rightarrow$ engaged
	`pitch_threshold`	[5°, 40°]	$\lvert\phi\rvert <$ this $\Rightarrow$ engaged
Posture event detectors	`face_cover_ratio`	[0.3, 0.9]	mouth-cover fires if ratio $<$ this
	`withdrawal_threshold_m`	[0.05, 0.30]	torso Δz for withdrawal (m)
	`posture_drop_gate`	[0.10, 0.25]	withdrawal gated by posture drop
Posture event penalties	`mouth_cover_penalty`	[0.0, 50.0]	points subtracted on mouth-cover
	`withdrawal_penalty`	[0.0, 50.0]	points subtracted on withdrawal
Temporal smoothing	`ema_time_constant_s`	[0.1, 2.0]	EMA τ, FPS-independent (s)
Missing-detection EMA decay	`no_face_target`	[30, 80]	emotion decay target, face missing
	`no_face_rate`	[0.05, 0.50]	decay rate, face missing (1/s)
	`no_pose_target`	[30, 80]	posture decay target, pose missing
	`no_pose_rate`	[0.05, 0.50]	decay rate, pose missing (1/s)

Why these groupings matter. Three of the seven groups carry most of the empirical weight. The *phase-aware fusion weights* (rows 1–6) determine which sub-score dominates in each interaction phase; the no-pose ablation discovery is literally the observation that `wp_intent` and `wp_exec` calibrate cleanly to zero under our current domain shift. The *gaze cone thresholds* widened from the default 25°/20° to roughly 32°/30° in the deployed config — the change that let `sc02`'s gaze signal stay positive while users reached for the cup. The *missing-detection EMA decay* group (rows 17–20) was hardcoded before this work; promoting these four constants to tunables was the single change that flipped the inverted `sc01`/`sc02` ordering, by raising `no_face_rate` from 0.15/s to roughly 0.47/s and dragging `sc01`'s sparse face-detection frames toward the low end of the comfort range.

What is *not* in the search space. The pretrained perception models are frozen — we do not fine-tune RetinaFace, EfficientNet-B0, L2CS-Net, or MediaPipe Pose. The `approach`-phase fusion weights are also fixed at defaults (w_e=0.5, w_p=0.5, w_g=0.3) because the approach window contributes only a small fraction of frames per recording and adding three more dimensions for marginal gain was not justified by the information density of the dataset. The abort threshold τ^* is selected post-hoc from the test-set ROC sweep; including it inside DE would conflate calibration with operating-point selection.

Optimization Objective

The calibration searches for a parameter vector $\theta \in \mathbb{R}^{20}$ (Table 2) that minimizes the per-fold cross-validated loss

$$\mathcal{L}(\theta) \;=\; -\Big[\,\overline{J}(\theta) \;+\; \varepsilon \cdot \frac{\mu_{\mathrm{sc02}}(\theta) - \mu_{\mathrm{sc01}}(\theta)}{100}\,\Big], \tag{1}$$

with $\varepsilon = 0.1$. Equation (1) pairs a primary classification term $\overline{J}(\theta)$ with a small linear margin tie-breaker. The two terms are derived below.

Per-recording reduction. For each training recording i, the runtime pipeline is replayed under parameters θ, producing a per-frame integrated comfort series $\tilde{C}(t;\theta)$. The recording is summarized by the mean over its intent + execution window W_i (defined by the sidecar JSON labels):

$$\bar{C}_i(\theta) \;=\; \frac{1}{|W_i|} \sum_{t \in W_i} \tilde{C}(t;\theta).$$

Cross-validated Youden's J. The training set is split into K=5 label-stratified folds. For each held-out fold f, the threshold τ is swept over [30, 80] and Youden's J is taken at its fold-wise maximum:

$$J_f(\theta) \;=\; \max_{\tau \in [30,\,80]} \left(\frac{\mathrm{TP}_f(\tau)}{n_f^{\mathrm{abort}}} + \frac{\mathrm{TN}_f(\tau)}{n_f^{\mathrm{cont}}} - 1 \right),$$

where $\mathrm{TP}_f(\tau)$ counts abort-labelled recordings in fold f with $\bar{C}_i \le \tau$ and $\mathrm{TN}_f(\tau)$ counts continue-labelled recordings with $\bar{C}_i > \tau$. The cross-validated term is the mean across folds:

$$\overline{J}(\theta) \;=\; \frac{1}{K} \sum_{f=1}^{K} J_f(\theta).$$

Scenario margin (tie-breaker). $\mu_{\mathrm{sc02}}(\theta)$ and $\mu_{\mathrm{sc01}}(\theta)$ are the means of $\bar{C}_i(\theta)$ over all training recordings labelled with each scenario, across all folds. The margin term explicitly rewards separation between the representative continue scenario (`sc02`, comfortable handoff) and the representative abort scenario (`sc01`, walk-by), independent of where the optimal threshold lands. The weight $\varepsilon = 0.1$ makes this a secondary signal; the primary optimization pressure comes from $\overline{J}$.

Optimizer. Equation (1) is minimized using SciPy's `differential_evolution` with population multiplier popsize=15 (yielding $15 \times 20 = 300$ candidates per generation), maxiter=50, convergence tolerance 10^{-3}, fixed seed 42, deferred whole-generation updating, and a final L-BFGS-B polishing pass. Differential evolution is gradient-free, which is required because $\overline{J}$ is a rank-based classification metric and non-differentiable in θ.

Variable summary.

Reading the loss in plain terms. Equation (1) says: *find parameters so that some threshold cleanly separates aborts from continues on the per-recording mean comfort (Youden's J), and as a tie-breaker, make the comfortable-handover scenario score higher than the walk-by scenario.* Trajectory shape (whether the score trends up or down within a recording) is not in this loss — it was

Symbol	Meaning
$\theta \in \mathbb{R}^{20}$	parameter vector (Table 2)
$\tilde{C}(t;\theta)$	per-frame integrated comfort score, $\in$ [0, 100]
W_i	intent + execution window of recording i (sidecar-derived)
$\bar{C}_i(\theta)$	per-recording mean of $\tilde{C}$ over W_i
τ	abort threshold, swept in [30, 80] within each fold
$J_f(\theta)$	Youden's J on fold f at the fold-optimal τ
$\overline{J}(\theta)$	mean Youden's J across the K=5 CV folds
$\mu_{\text{sc02}}, \mu_{\text{sc01}}$	training-set scenario means of $\bar{C}_i$
ε	margin weight = 0.1 (tie-breaker)
K	number of stratified CV folds = 5

an evaluation gate against which the final calibrated parameters were inspected after the search. The shape-aware objective variants F, G, H, and I that replaced $\overline{J}$ with tanh-encoded slope terms all reduced classification accuracy and lost the variant race; Variant A (Eq. 1) remained the winner.